Derivatives: Contracts, Models, and Markets · Technical Note 36

The Samuelson Maturity Effect

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Section 36.3 of Derivatives: Contracts, Models, and Markets states that when the spot price of a commodity mean-reverts, the volatility of a futures price falls with the contract's distance to delivery as $\sigma_S e^{-a(T-t)}$, and it states this without proof. This note derives it from a mean-reverting log spot price, in four steps that each do one thing: solve the spot's dynamics, take the expectation that defines a futures price, apply Itô's lemma, and use the fact that a futures price is a martingale. It then compares the result with Hull–White's forward-rate volatility, which has the same decay and measures a different quantity; derives the average variance that Black's formula needs for an option on a futures contract, which depends on two maturities rather than one; and shows why the same model makes every contract month perfectly correlated, which is the reason the chapter calls for a second factor as soon as a spread is involved.

`tools/verify/tn36-the-samuelson-effect.py` *recomputes every result here.*

1 THE MODEL

Let $X(t) = \ln S(t)$ be the logarithm of the spot price, and suppose that under the pricing measure Q it reverts toward a level μ:

$$dX = a\,(\mu - X)\,dt + \sigma_S\,dW. \tag{TN36.1}$$

This is the one-factor commodity model of Schwartz (1997). Two of its symbols need fixing before anything is derived.

σ_S is the spot volatility in the ordinary sense. By Itô's lemma $dS/S = (\cdots)\,dt + \sigma_S\,dW$, so σ_S is the volatility of the spot's percentage return, which is how the chapter's 60% is meant.

μ is a risk-adjusted level. It absorbs the convenience yield of Section 36.1 and any market price of risk, and it will drop out of every volatility below. That is why the chapter's result contains no μ, and why the result can be used without estimating one.

2 THE LOG PRICE AT DELIVERY

Equation (TN36.1) is an Ornstein–Uhlenbeck process, and the integrating factor e^{as} solves it:

$$d\big(e^{as}X(s)\big) = e^{as}\,dX + a\,e^{as}X\,ds = a\mu\,e^{as}\,ds + \sigma_S\,e^{as}\,dW(s).$$

Integrating from t to T and writing $\tau = T - t$,

$$X(T) = e^{-a\tau}X(t) + \mu\big(1 - e^{-a\tau}\big) + \sigma_S\int_t^T e^{-a(T-s)}\,dW(s). \tag{TN36.2}$$

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Given what is known at t, the stochastic integral is normal with mean zero and, by the Itô isometry, variance

$$V(\tau) = \sigma_S^2 \int_t^T e^{-2a(T-s)}\, ds = \frac{\sigma_S^2}{2a}\big(1 - e^{-2a\tau}\big). \tag{TN36.3}$$

So $X(T)$ is normal, and today's log price enters it with weight $e^{-a\tau}$. *A shock to today's spot survives to the delivery date only in that proportion.* The factor is the whole result; the remaining steps carry it from the spot to the futures price.

3 THE FUTURES PRICE

A futures price is the Q-expectation of the spot at delivery, $F(t,T) = E_t^Q[S(T)] = E_t^Q\big[e^{X(T)}\big]$. For a normal variable $E[e^Y] = e^{\,m+v/2}$, where m and v are its mean and variance, so from (TN36.2) and (TN36.3)

$$\ln F(t,T) = e^{-a\tau}X(t) + \mu\big(1 - e^{-a\tau}\big) + \frac{\sigma_S^2}{4a}\big(1 - e^{-2a\tau}\big). \tag{TN36.4}$$

Only the first term is random. The other two are deterministic functions of the time to delivery.

4 THE VOLATILITY OF THE FUTURES PRICE

Apply Itô's lemma to (TN36.4), holding T fixed as t moves, so that $d\tau = -dt$. The first term gives

$$d\big(e^{-a\tau}X\big) = a\,e^{-a\tau}X\,dt + e^{-a\tau}\big[a(\mu - X)\,dt + \sigma_S\,dW\big] = a\mu\,e^{-a\tau}\,dt + \sigma_S\,e^{-a\tau}\,dW,$$

the second gives $-a\mu\,e^{-a\tau}\,dt$, and the third gives $-\frac{1}{2}\sigma_S^2 e^{-2a\tau}\,dt$. The two μ terms cancel, which is the promised disappearance of μ, and what remains is

$$d\ln F(t,T) = -\frac{1}{2}\sigma_S^2 e^{-2a(T-t)}\,dt + \sigma_S\,e^{-a(T-t)}\,dW. \tag{TN36.5}$$

The drift is exactly $-1/2$ times the square of the diffusion coefficient, which is what a driftless dF/F looks like after the passage to logarithms. That is the check that F is a Q-martingale, as a futures price must be, and it confirms

$$\frac{dF(t,T)}{F(t,T)} = \sigma_S\,e^{-a(T-t)}\,dW,$$

so that the futures volatility is

$$\sigma_F(t,T) = \sigma_S\,e^{-a(T-t)}, \tag{TN36.6}$$

which is the chapter's equation (36.2).

Three limits confirm the shape. At $T = t$ the futures price is the spot price and $\sigma_F = \sigma_S$. As $T - t \to \infty$ the volatility of the futures price vanishes, because a shock today has decayed before delivery. At $a = 0$ the log spot is a random walk, every contract carries the spot's volatility, and the model is the Black–Scholes–Merton one.

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Example 4.1 The table in Example 36.2

At $\sigma_S = 60\%$ and $a = 1.5$, (TN36.6) gives

$$\sigma_F(1 \text{ month}) = 0.60\, e^{-1.5/12} = 52.9\%,$$
$$\sigma_F(12 \text{ months}) = 0.60\, e^{-1.5} = 13.4\%,$$

and the ratio of the two is $e^{1.5\times 11/12} = 3.96$, the factor the chapter quotes.

5 COMPARISON WITH HULL–WHITE

The chapter notes that the same decay appears in Section 31.6. There the *short rate* is the Ornstein–Uhlenbeck process, $dr = [\theta(t) - ar]\,dt + \sigma\,dW$, and differentiating the affine bond price gives the instantaneous forward rate as

$$f(t,T) = (\text{deterministic}) + e^{-a(T-t)}\, r(t),$$

so that $df(t,T) = (\cdots)\,dt + \sigma\, e^{-a(T-t)}\,dW$. Seen from today, $t = 0$, this is σe^{-aT}, equation (31.7) of Chapter 31.

	Hull–White, (31.7)	**Samuelson, (36.2)**
Mean-reverting variable	short rate r	log spot price $\ln S$
Quantity measured	df, a change in a rate	dF/F, a percentage change in a price
Units of the volatility	basis points	percent
Distribution at the long date	normal forward rate	lognormal futures price

The derivation is the same in both: the quantity at the long date carries today's state with weight $e^{-a(T-t)}$, and its volatility inherits the weight. What differs is which variable is Gaussian, and so whether the decaying volatility is an absolute one or a percentage one. The two numbers are therefore not interchangeable: a 100 bp Hull–White σ and a 60% commodity σ_S describe different objects even when they decay at the same a.

6 THE IMPLIED VOLATILITY OF AN OPTION ON A FUTURES CONTRACT

An option expiring at T_0 on the futures contract for delivery at $T \geq T_0$ is priced by Black's formula, which needs the variance of $\ln F(T_0,T)$ accumulated over the option's life. By (TN36.5) the instantaneous volatility over $0 \leq t \leq T_0$ is $\sigma_S e^{-a(T-t)}$, so the average variance is

$$\bar{\sigma}^2 = \frac{1}{T_0}\int_0^{T_0} \sigma_S^2\, e^{-2a(T-t)}\,dt = \sigma_S^2\, e^{-2a(T-T_0)}\,\frac{1-e^{-2aT_0}}{2aT_0}. \qquad \text{(TN36.7)}$$

The same number follows by a second route, without the integral: by (TN36.4) the random part of $\ln F(T_0,T)$ is $e^{-a(T-T_0)}X(T_0)$, and (TN36.3) gives the variance of $X(T_0)$ seen from today, so

$$\operatorname{Var}_0\big[\ln F(T_0,T)\big] = e^{-2a(T-T_0)}\,\frac{\sigma_S^2}{2a}\big(1-e^{-2aT_0}\big) = \bar{\sigma}^2\, T_0.$$

The implied volatility therefore depends on two maturities, the option's and the futures contract's. When the two coincide, $T_0 = T$, it reduces to $\sigma_S\sqrt{(1-e^{-2aT})/(2aT)}$,

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which still falls with maturity; when the futures contract runs past the option, the factor $e^{-a(T-T_0)}$ lowers it further. That is the precise content of the chapter's remark that a commodity volatility surface needs a maturity dimension that means something different from an equity surface's.

Example 6.1 An option on a later futures contract

Take Example 36.2's $\sigma_S = 60\%$ and $a = 1.5$, and an option expiring in nine months on the twelve-month futures contract: $T_0 = 0.75$, $T = 1$. Then

$$\bar{\sigma}^2 = 0.36\, e^{-2\times1.5\times0.25}\,\frac{1 - e^{-2\times1.5\times0.75}}{2\times1.5\times0.75} = 0.36\times0.4724\times0.3976 = 0.0676,$$

an implied volatility of **26.0%**. The futures price's instantaneous volatility runs from 13.4% when the option is written to 41.2% when it expires, and the option prices on the root-mean-square of that path. Quoting it at either end would misprice it badly, at 13.4% by half and at 41.2% by more than half again.

7 WHY ONE FACTOR MAKES EVERY CONTRACT MONTH PERFECTLY CORRELATED

Every futures price in (TN36.6) is driven by the same dW, with a loading $\sigma_S e^{-a(T-t)}$ that differs by contract but never changes sign. The instantaneous correlation between any two contract months is therefore

$$\rho\big(dF_1/F_1,\ dF_2/F_2\big) = \frac{\sigma_S^2\, e^{-a(T_1-t)}\, e^{-a(T_2-t)}}{\sigma_S e^{-a(T_1-t)}\ \sigma_S e^{-a(T_2-t)}} = 1.$$

The model lets near and far contracts move by different amounts, which is the Samuelson maturity effect, but always in the same direction and in fixed proportion. A calendar spread hedged in that proportion is riskless, and an option on the spread is worth only what the difference between the two contracts' volatilities gives it, which is little once the model is fitted to a realistic long end; Example 36.7 of the chapter puts it at a twelfth of the two-factor value. This is the one-factor defect of Section 31.12 in another market, and the chapter's remedy is the same one: a second factor, usually a random walk in the long-run level μ, which moves the far end of the curve independently of the near end and restores a correlation below one.

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